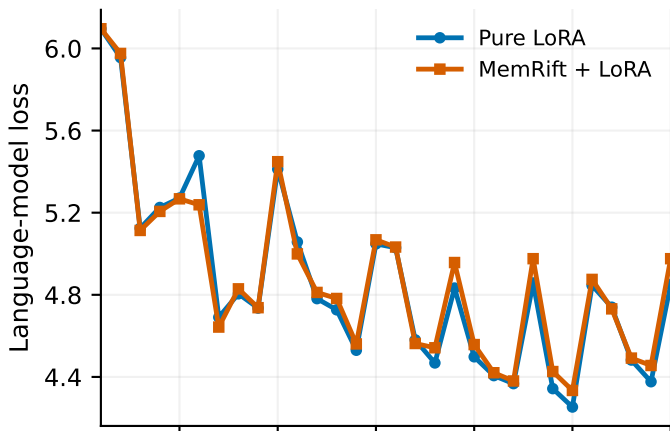
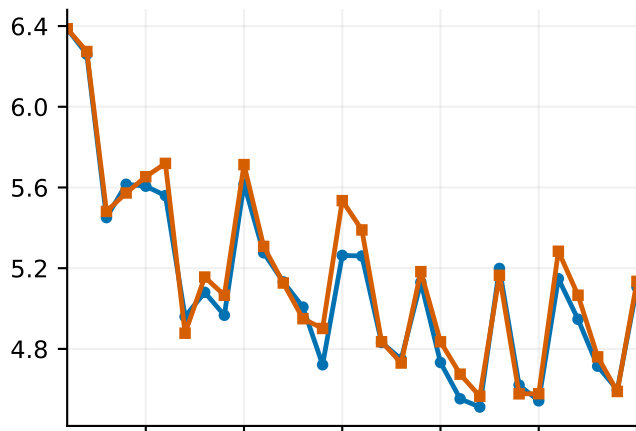


First 30 Training Steps: Loss Agreement under atol=1e-2, rtol=2e-3

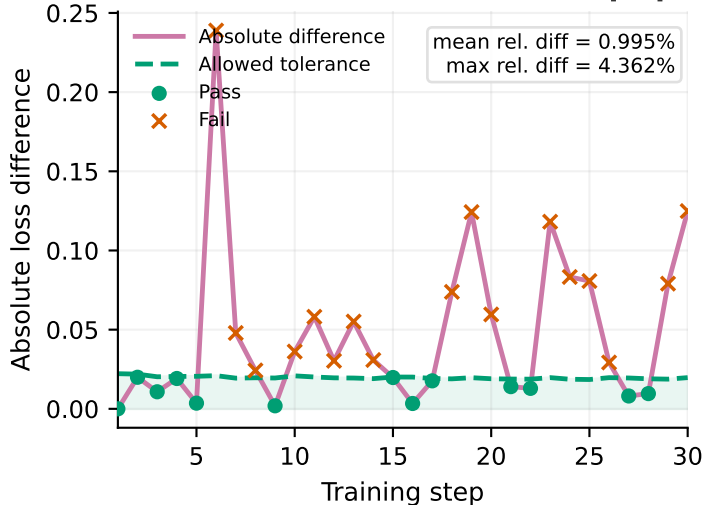
(a) LLaMA-3.1-8B: loss



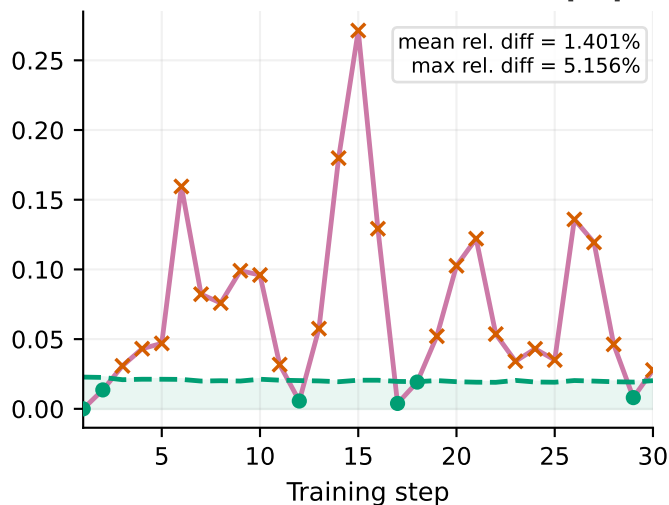
(b) Aquila2-7B: loss



(c) Absolute difference: 13/30 steps pass



(d) Absolute difference: 6/30 steps pass



$$\text{Pass criterion: } |L_{\text{MemRift}} - L_{\text{Pure}}| \leq 10^{-2} + 2 \times 10^{-3} |L_{\text{Pure}}|.$$